

QUIDDITY INNOVATIONS

QI Ecosystem — Technology Stack Reference

An inverted, technology-first view of every tool in the ecosystem

What each technology is · what it is used for · which applications rely on it

22 applications · 141 technologies · 14 categories

Generated 2026-06-27 · Source: QI Hive Project-Status library

How to read this document

Most technology documentation is written application-first: it lists one app and the stack behind it. This reference inverts that perspective. It starts from the technologies themselves — the languages, frameworks, models, drivers and services that the Quiddity Innovations ecosystem is built on — and, for each one, answers three questions:

- What it is — a short, vendor-neutral explanation of the technology and its licence.
- What QI uses it for — the concrete jobs it does across the ecosystem.
- Which applications use it — the full list of QI tools that depend on it.

Technologies are grouped into the categories below. Within each category they are ordered by how many applications share them, so the load-bearing, ecosystem-wide technologies appear first and the specialist, single-application components appear last.

Categories

1. **Languages & Runtimes** *(10 technologies)*
2. **Web & API Frameworks / Servers** *(6 technologies)*
3. **Frontend, UI & Styling** *(8 technologies)*
4. **Data Validation, HTTP Clients & Concurrency** *(8 technologies)*
5. **Databases, Vector Stores & On-Disk State** *(9 technologies)*
6. **AI — Local Inference (LLMs, Embeddings, Speech, Vision, Media)** *(19 technologies)*
7. **AI — Cloud & Hosted Models** *(11 technologies)*
8. **Document, PDF, Office & Media Processing** *(20 technologies)*
9. **Service Management, Tunnels & Scheduling** *(9 technologies)*
10. **External Integrations & Data Sources** *(14 technologies)*
11. **Security, Auth & Cryptography** *(6 technologies)*
12. **Developer Tooling, Agents & Protocols** *(9 technologies)*
13. **QI Internal Platform Services** *(6 technologies)*
14. **Other / Project-Specific Components** *(6 technologies)*

The shared backbone — technologies used across the ecosystem

These are the technologies that recur across the most applications. They define the common shape of a QI tool: a Python service, validated with Pydantic, served by FastAPI/Uvicorn or the standard library, talking to SQLite, reasoning with a local Ollama model, generating Word docs with python-docx, and kept alive by an NSSM service behind a Cloudflare tunnel.

Technology	Apps	Category
Python (CPython)	21	Languages & Runtimes
NSSM (Windows service manager)	18	Service Management, Tunnels & Scheduling
FastAPI	15	Web & API Frameworks / Servers
Ollama	15	AI — Local Inference (LLMs, Embeddings, Speech, Vision, Media)
Uvicorn	14	Web & API Frameworks / Servers
SQLite	12	Databases, Vector Stores & On-Disk State
JSON files as state store	12	Databases, Vector Stores & On-Disk State
Pydantic	11	Data Validation, HTTP Clients & Concurrency
httpx	10	Data Validation, HTTP Clients & Concurrency
Anthropic Claude (API / SDK)	9	AI — Cloud & Hosted Models
Gradio	7	Web & API Frameworks / Servers
JavaScript (ES Modules / vanilla)	7	Languages & Runtimes
requests	7	Data Validation, HTTP Clients & Concurrency
APScheduler	5	Service Management, Tunnels & Scheduling

Languages & Runtimes

Python (CPython)

Licence: Apache 2.0, Apache 2.0 / Apache 2.0, BSD, MIT, MIT / Apache 2.0, Open source (BSD), PSF, PSF (open source), PSF (stdlib), PSF / MIT · Used by 21 applications

What it is. The reference implementation of the Python programming language. A general-purpose, dynamically-typed language with a vast standard library; the dominant choice for back-end services, data work and AI tooling. Open-source under the PSF license.

How QI uses it. • Sole implementation language for the API, Gradio UI, scout engine, adapters, and shared modules • Sole implementation language for the Gradio app, pipeline, TTS, render wrappers, and scene engine • All hooks, the supervisor, the self-audit, reconciliation + migration scripts, and the docx generators. • Sole implementation language for the control API, channels, realtime loop, meeting room, and backends

Applications. AkiyaScout, AvatarStudio, Claude Manager, Claude Voice, CogniBase, CypherMiner, Digitization Cost Tool, EasyFlow, FileHQ, Gamez, LotteryWiz, M2V, MQ, MapSnap, NEXUS, Naya, OpenClaw, PersonalSong Studio, QI Brain, QI Hive, TubeScout

JavaScript (ES Modules / vanilla)

Licence: Built-in (ES2020), Open standard, Proprietary (QI), Proprietary (Quiddity Innovations), n/a (app code), n/a (in-repo) · Used by 7 applications

What it is. The native scripting language of the browser. Used here in modern ES-module form with no framework, keeping front-ends dependency-free and portable.

How QI uses it. • All UI logic in app.js (~5,600 lines); no framework, no bundler, no jQuery. • The entire UI is hand-built with a small el() helper in core/ui.ts and generated from the tool registry — no React/Vue/Svelte • Arbitrary-precision integers for factorization, combinatorics, lottery odds and large-sequence math • All calculation, rendering, persistence, export/import, and pagination. Zero runtime dependencies, no framework, no bundler.

Applications. AutoPDF, CypherMiner, Digitization Cost Tool, EasyFlow, LotteryWiz, MapSnap, TubeScout

Windows Batch (.bat)

Licence: Microsoft (OS), Windows · Used by 4 applications

What it is. The classic Windows command-script format (.bat/.cmd), used for double-clickable launchers and Explorer-invoked wrappers.

How QI uses it. • Current launcher — cd C:\AkiyaScout && python main.py. NSSM QI_AkiyaScout service is planned, not yet registered • Explorer-launched Clean-Restart wrapper (run outside a Claude session for a true clean restart). • dist/ holds the packaged build: index.html + User Guide + Deployment Guide, zipped for hand-off to Document Imaging & Services. • Preferred service start/stop (nssm OpenService access-denied workaround)

Applications. AkiyaScout, Claude Manager, Digitization Cost Tool, QI Hive

PowerShell

Licence: BSD, MIT (PowerShell Core) / Windows, MIT (PowerShell) / proprietary (Windows PowerShell 5.1) · Used by 3 applications

What it is. Microsoft's object-oriented command shell and scripting language for Windows automation — service control, process management and lightweight HTTP servers (HttpListener).

How QI uses it. • HTTP server (HttpListener) on 127.0.0.1:6969 + the entire processing engine; hosts static UI and ~80 /api/* routes. • Restart guard, monthly-audit orchestrator, service-rights grant, and the supervisor's service/scheduled-task probes. • Mode-A push-to-talk: transcript to clipboard or auto-typed into the focused window

Applications. AutoPDF, Claude Manager, Claude Voice

Bash / Shell

Licence: GPL · Used by 1 application

What it is. The Unix/Linux command shell and scripting language, used for orchestration glue, health checks and scheduled entry points. GPL.

How QI uses it. • Orchestration shells for digests, briefings, health checks, and Task Scheduler entry points.

Applications. OpenClaw

C# / .NET

Licence: Proprietary (Quiddity Innovations) · Used by 1 application

What it is. A statically-typed, compiled language and runtime from Microsoft. csc.exe ships with Windows, so small C# launchers can be compiled with zero external SDK install. Proprietary toolchain, runtime free to use.

How QI uses it. • AutoPDF.exe idempotent launcher (Launcher.cs) compiled with Windows built-in csc.exe; embedded .ico, splash.

Applications. AutoPDF

Google Apps Script (V8)

Licence: Free (Google quotas apply) · Used by 1 application

What it is. Google's cloud-hosted JavaScript (V8) runtime that executes inside a user's Google account, used to run server-side email classifiers on a schedule. Free within Google quotas.

How QI uses it. • Server-side classifier generated from config: classifyNewEmails (15-min), triageStaleInbox (daily 1 AM), TIER_RULES engine — runs on Google's servers, no PC required

Applications. EasyFlow

Node.js

Licence: MIT-style · Used by 1 application

What it is. A server-side JavaScript runtime built on Chrome's V8 engine; here it hosts a lightweight static dashboard server. MIT.

How QI uses it. • Lightweight static dashboard server (agents, tasks, roadmap, sentry status) on 127.0.0.1:18800.

Applications. OpenClaw

OpenJDK / Java JRE

Licence: GPL-2.0-with-classpath-exception · Used by 1 application

What it is. A free, open-source Java runtime (OpenJDK / Liberica) bundled to run Java-based engines such as Tabula and pdftk-java without a system Java install. GPL+CE.

How QI uses it. • Bundled JRE that runs the Java engines (Tabula, pdftk-java).

Applications. AutoPDF

TypeScript

Licence: Apache 2.0 · Used by 1 application

What it is. A typed superset of JavaScript that compiles to plain JS, adding compile-time type checking and IDE tooling to front-end and tool code. Apache-2.0, by Microsoft.

How QI uses it. • Sole implementation language for the tool suite — every tool, the registry, UI shell, i18n, assistant and visuals

Applications. CypherMiner

Web & API Frameworks / Servers

FastAPI

Licence: MIT, MIT / BSD · Used by 15 applications

What it is. A modern, high-performance Python web framework for building REST APIs, with automatic validation and OpenAPI docs. The default API layer across the QI ecosystem. MIT.

How QI uses it. • REST API on port 8505 — QI /health//version//info contract plus listings/profiles/ingest/notifications/digest/regions/sources • Loopback control/health API + brain config UI on :8720 (/health, /version, /info, /config, /brain*, /bridge/health, /say) • REST API on port 8650 — /health, /chat, /vendors, /anchors, /capabilities, /visualizer (6 mounted routers) • REST API on port 8502 — /health, /version, /info (QI contract); CORS from the QI registry

Applications. AkiyaScout, Claude Voice, CogniBase, CypherMiner, FileHQ, Gamez, LotteryWiz, M2V, MQ, NEXUS, Naya, PersonalSong Studio, QI Brain, QI Hive, TubeScout

Uvicorn

Licence: BSD · Used by 14 applications

What it is. A lightning-fast ASGI server that runs FastAPI applications. The standard process that actually serves QI's FastAPI apps. BSD.

How QI uses it. • ASGI server running the FastAPI app (started from main.py on a daemon thread) • ASGI server hosting the FastAPI apps (server.py :8720, line_bot.py :8721, meeting_server.py :8722), log_level=warning • ASGI server hosting the FastAPI app (run() entry point / cognibase-server script) • ASGI server running the FastAPI app (run_api.py / main.py)

Applications. AkiyaScout, Claude Voice, CogniBase, CypherMiner, FileHQ, Gamez, LotteryWiz, M2V, MQ, NEXUS, PersonalSong Studio, QI Brain, QI Hive, TubeScout

Gradio

Licence: Apache 2.0 · Used by 7 applications

What it is. An open-source Python library that turns scripts into shareable web UIs with almost no front-end code; QI uses it for operator/admin dashboards. Apache-2.0.

How QI uses it. • Operator dashboard on port 7845 — 6 tabs (Map / Discover / Profiles / Matches / Notifications / Settings) with an EN/日本語/Both toggle and a dark slate theme • Studio web UI on port 7862 — agent picker, per-language scripts, engine config, live progress, video preview • Single-page operator console on port 7841 — upload, style picker, reference photo, live progress log, video player (purple/zinc Base theme) • Operator console on port 7840 — minimal Status + About shell, pings the API health endpoint

Applications. AkiyaScout, AvatarStudio, M2V, MQ, NEXUS, Naya, TubeScout

Jinja2 templating

Licence: BSD, BSD / open standard · Used by 2 applications

What it is. A Python templating engine that renders HTML server-side from templates and data — the basis of several server-rendered dashboards. BSD.

How QI uses it. • 11 templates (base, dashboard, configure, configure_tier, designer, deploy, cleanup, migrate, settings, welcome, manual) served by Flask • Server-rendered dashboard — 11 templates (dashboard, duplicates, explorer, organizer, tasks, history, reports, exceptions, settings, structure, base)

Applications. EasyFlow, FileHQ

Caddy

Licence: Apache-2.0 · Used by 1 application

What it is. A modern open-source web server / reverse proxy with automatic HTTPS, used as a LAN front door with an internal certificate authority. Apache-2.0.

How QI uses it. • LAN reverse proxy — qi.local front door with internal-CA HTTPS (QI_Caddy)

Applications. QI Hive

Flask

Licence: BSD · Used by 1 application

What it is. A minimal, widely-used Python web microframework — simpler than FastAPI, used where a small synchronous HTTP server is enough. BSD.

How QI uses it. • HTTP server on port 8002 — /ask, /health, /status, /version, /info, /api/action, /webhook/*

Applications. Naya

Frontend, UI & Styling

Single-file / single-page HTML-CSS-JS

Licence: Proprietary (QI), Proprietary (Quiddity Innovations) · Used by 4 applications

What it is. A self-contained front-end built as one (or few) hand-written HTML files with inline CSS and vanilla JavaScript — no build step, no framework, maximally portable. Proprietary QI source.

How QI uses it. • AutoPDFTool.html single-page shell (7 tabs) + hand-written styles.css; zone-mapper.html separate mapper tool. • Dark/light theming via a data-theme attribute; layout in src/styles.css • The entire tool — UI, cost model, report generator — is one self-contained index.html (~1,450 lines). No build step required to run it. • BU-branded layout (CSS custom properties for the BU red palette), responsive input grids, bar charts, and a print stylesheet that paginates the report to letter size.

Applications. AutoPDF, CypherMiner, Digitization Cost Tool, Gamez

Browser storage (localStorage / Web Storage)

Licence: Browser built-in, Web standard · Used by 2 applications

What it is. The browser's built-in key-value store, used to persist user preferences and small state entirely client-side with no server. Web standard.

How QI uses it. • Stores language, theme, favourites, sidebar pins and assistant-open state — no server, no cookies • Stores rate table (budct_rates_v2), last inputs (budct_inputs_v2), preparers, password hash, and prefs — entirely client-side, no server.

Applications. CypherMiner, Digitization Cost Tool

AdminLTE

Licence: MIT · Used by 1 application

What it is. A free Bootstrap-based admin dashboard theme (sidebar, cards, tables, badges). MIT.

How QI uses it. • Layout + component library (sidebar, cards, badges, tables)

Applications. QI Hive

Bootstrap

Licence: MIT · Used by 1 application

What it is. The most widely-used open-source CSS framework for responsive layouts and components. MIT.

How QI uses it. • CSS utility framework underlying AdminLTE • bi-* icon set used throughout the dashboard and the status tabs

Applications. QI Hive

folium / Leaflet

Licence: MIT · Used by 1 application

What it is. folium is a Python library that renders interactive Leaflet.js maps (choropleths, clustered markers, base-layer switching) from data. MIT.

How QI uses it. • Interactive Leaflet map of Japan — prefecture choropleth, clustered price pins, base-layer + hazard-overlay toggles

Applications. AkiyaScout

pywebview

Licence: BSD · Used by 1 application

What it is. A Python library that opens a web UI inside a native OS window (Edge WebView2 / WebKit) instead of a browser tab, for a desktop-app feel. BSD.

How QI uses it. • Opens the app in a native OS webview window (Edge WebView2 / WebKit / WebKitGTK) instead of a browser tab

Applications. Gamez

SortableJS

Licence: MIT · Used by 1 application

What it is. A small JavaScript library for drag-and-drop lists and kanban boards. MIT.

How QI uses it. • Drag-and-drop kanban board between columns

Applications. QI Hive

Vite

Licence: MIT · Used by 1 application

What it is. A fast modern front-end build tool / dev server that bundles TypeScript/JS for production. MIT.

How QI uses it. • Dev server (port 7842) and static production bundler (tsc && vite build -> frontend/dist) • Generates the service worker so the suite is installable and works fully offline

Applications. CypherMiner

Data Validation, HTTP Clients & Concurrency

Pydantic

Licence: MIT · Used by 11 applications

What it is. The de-facto Python data-validation library; defines request/response models with type enforcement and powers FastAPI's validation. MIT.

How QI uses it. • Request models — ProfileIn / ProfilePatch validation for the profiles API • Request models (BrainSet, SayBody) for the control endpoints • Request models for chat, anchors, and capability CRUD/run/grant/draft/validate • Request/response validation for FastAPI (pulled in for future endpoints)

Applications. AkiyaScout, Claude Voice, CogniBase, CypherMiner, M2V, MQ, NEXUS, PersonalSong Studio, QI Brain, QI Hive, TubeScout

httpx

Licence: BSD · Used by 10 applications

What it is. A modern Python HTTP client supporting both sync and async requests — used for polite crawling, geocoding and calling other services. BSD.

How QI uses it. • Synchronous HTTP client for polite rate-limited crawling, geocoding, DeepL/Ollama translation, and detail-page enrichment • Sync HTTP client for the OnBase REST client (and Unity REST overlap) • Used by the legacy Gradio placeholder to poll the API; not part of the real tool suite • Sync fetches of ESPN/API-Football/Kalshi (sources.py) and async relays to OpenRouter/Ollama (server.py)

Applications. AkiyaScout, CogniBase, CypherMiner, Gamez, LotteryWiz, MQ, NEXUS, PersonalSong Studio, QI Brain, TubeScout

requests

Licence: Apache 2.0, MIT / Apache 2.0, Telegram ToS · Used by 7 applications

What it is. The classic, ubiquitous Python HTTP library for simple synchronous API calls and file downloads. Apache-2.0.

How QI uses it. • D-ID / HeyGen REST uploads, generate, status polling, result download • UI -> API calls and the event-polling loop; also used by the lyric analyzer to call Ollama • HTTP client used by the smoke tests; declared in requirements • HTTP client to Ollama, OpenRouter, Cloudflare, NEXUS /synthesize, and Telegram getUpdates

Applications. AvatarStudio, Claude Voice, FileHQ, M2V, MQ, Naya, TubeScout

asyncio

Licence: MIT, PSF (stdlib) · Used by 3 applications

What it is. Python's built-in asynchronous I/O framework, used to fan out many provider/API calls concurrently (asyncio.gather). PSF.

How QI uses it. • ~95 tests against FakeChatVendor/FakeEmbedVendor — no API keys required to run the suite • Concurrent fan-out: all providers dispatched simultaneously via asyncio.gather / as_completed • Async endpoints + the MCP stdio loop; run_in_executor avoids the Windows ProactorEventLoop pipe crash

Applications. CogniBase, NEXUS, QI Brain

aiofiles

Licence: Apache 2.0 · Used by 2 applications

What it is. Async file I/O for non-blocking reads/writes inside FastAPI/asyncio apps. Apache-2.0.

How QI uses it. • Async file I/O for the FastAPI layer • Async file I/O for uploads and exports

Applications. FileHQ, NEXUS

urllib (stdlib)

Licence: PSF (stdlib) · Used by 2 applications

What it is. Python's built-in URL/HTTP library, a zero-dependency fallback for fetching feeds and calling local services. PSF.

How QI uses it. • live_sources.py fetchers for data.ny.gov, Caixa, FDJ and community feeds • Fumi's actual HTTP calls — Cloudflare Workers AI, Ollama, and the Facebook Graph API are all called via urllib.request, not an external client

Applications. LotteryWiz, MQ

humanize / rich

Licence: MIT · Used by 1 application

What it is. humanize formats sizes/dates for people; rich renders styled console output. MIT.

How QI uses it. • Human-readable file sizes/dates in the dashboard (humanize) and console output (rich)

Applications. FileHQ

threading

Licence: PSF (stdlib) · Used by 1 application

What it is. Python's standard threading module for background daemon workers (pollers, SSE streaming). PSF.

How QI uses it. • Poller runs as a daemon thread started at API import; reads project state files + git on an interval

Applications. QI Brain

Databases, Vector Stores & On-Disk State

JSON files as state store

Licence: Internal (QI), MIT, MIT (spec), n/a (QI internal), n/a (data) · Used by 12 applications

What it is. Plain, human-readable, git-friendly JSON files used as the source of truth instead of a database — config, sidecars, draw stores, job state. Data, no license.

How QI uses it. • autopdf-settings.json, recipes/*.json, templates/*.json, regex_library.json, scheduler-jobs.json; no database. • Single source for every project's path, ports, status, and relationships — read by the supervisor and both hooks for project inference. • All settings live-editable in config.json; secrets (LINE/Telegram tokens, CLAUDE_CODE_OAUTH_TOKEN) in a gitignored .env loaded by backends._load_env • data/brain_state.json (active brain), data/line_control.json (owner controls), data/bridge_health.json — no database

Applications. AutoPDF, Claude Manager, Claude Voice, CogniBase, FileHQ, Gamez, LotteryWiz, M2V, MQ, MapSnap, QI Brain, QI Hive

SQLite

Licence: Public domain, Public domain / Apache 2.0 · Used by 12 applications

What it is. A serverless, single-file SQL database embedded directly in the application — zero-admin, the default relational store across QI. Public domain.

How QI uses it. • Single-file bilingual store: prefectures, sources, raw_listings, listings, translations, profiles, matches, schedules, notifications, geocache, settings • qi_brain.db is the reconciliation + session target; the librarian queries its docs/doc_relationships catalog and the qi_docs Chroma collection. • Per-run sandbox: declared anchors' JSONL is loaded into an ephemeral :memory: DB (one table per anchor) and the capability SQL runs there with safe parameter binding • Single-file catalogue: files, duplicate_groups, duplicate_members, operations, tasks, task_runs, exceptions, scans, schema_info

Applications. AkiyaScout, Claude Manager, CogniBase, FileHQ, LotteryWiz, MapSnap, NEXUS, Naya, OpenClaw, QI Brain, QI Hive, TubeScout

ChromaDB

Licence: Apache 2.0, Apache-2.0 · Used by 5 applications

What it is. An open-source embedding (vector) database for semantic search and RAG; stores text embeddings on disk and retrieves by meaning. Apache-2.0.

How QI uses it. • Persistent on-disk vector store at Product/ONBASE/RAG_Index; main collection cognibase_main + per-anchor anchor_<id> collections (~40 demo embeddings) • Local per-profile .index/ for Layer 5 semantic retrieval; also a ChromaExtractor source. • Persistent vector store (collection nexus_knowledge, cosine HNSW) for RAG augmentation of answers • Semantic memory: PersistentClient at qi_memory/, 4 collections (qi_decisions/qi_features/qi_sessions/qi_docs), cosine HNSW

Applications. CogniBase, MapSnap, NEXUS, QI Brain, QI Hive

Excel COM as data mirror

Licence: Commercial (Microsoft) · Used by 1 application

What it is. Optional .xlsx mirror of an index produced via Microsoft Excel COM automation.

How QI uses it. • Optional XLSX mirror of the CSV index.

Applications. AutoPDF

oracledb (Oracle driver)

Licence: Apache-2.0 / UPL · Used by 1 application

What it is. Oracle's official Python driver, used in thin (no-client-install) mode for live Oracle extraction. Apache-2.0 / UPL.

How QI uses it. • OracleExtractor live extract + query execution (thin mode).

Applications. MapSnap

psycopg2 (PostgreSQL driver)

Licence: LGPL · Used by 1 application

What it is. The standard PostgreSQL database adapter for Python. LGPL.

How QI uses it. • PostgresExtractor live extract + query execution.

Applications. MapSnap

PyMySQL (MySQL driver)

Licence: MIT · Used by 1 application

What it is. A pure-Python MySQL/MariaDB client. MIT.

How QI uses it. • MySqlExtractor live extract + query execution.

Applications. MapSnap

pyodbc (SQL Server driver)

Licence: MIT-0 · Used by 1 application

What it is. A Python ODBC bridge for connecting to SQL Server (e.g. OnBase, Jenzabar) for read-only extraction and queries. MIT-0.

How QI uses it. • Live extract + read-only query execution against SQL Server (OnBase hsi schema, Jenzabar).

Applications. MapSnap

Qdrant

Licence: Apache-2.0 · Used by 1 application

What it is. An open-source vector database; here treated as an additional browsable schema/source. Apache-2.0.

How QI uses it. • QdrantExtractor — treat a Qdrant collection as a browsable 'schema' source.

Applications. MapSnap

AI — Local Inference (LLMs, Embeddings, Speech, Vision, Media)

Ollama

Licence: Apache 2.0, Apache-2.0, MIT, MIT (Ollama), MIT (runtime), MIT / mixed, MIT runtime / Apache 2.0 model, Open source models · Used by 15 applications

What it is. An open-source runtime that downloads and serves large-language models locally over an HTTP API (localhost:11434) — 100% private, free inference with no cloud. MIT.

How QI uses it. • Free local JA->EN fallback (default model qwen2.5:7b) — the engine behind all 187 cached translations today • Optional local LLM host (127.0.0.1:11434) for Smart Mapping chat, AI field proposal, and ai_extract fields. • Local, free, low-latency stand-in 'Ronald' (llama3.1:8b) on the RTX 5080; the always-on default brain • Local, free, private classifier/chat via /api/chat — no data leaves the machine

Applications. AkiyaScout, AutoPDF, Claude Voice, CogniBase, EasyFlow, Gamez, LotteryWiz, M2V, MQ, MapSnap, NEXUS, OpenClaw, PersonalSong Studio, QI Brain, QI Hive

faster-whisper / Whisper (STT)

Licence: MIT, MIT / Unlicense · Used by 5 applications

What it is. Local, free speech-to-text. faster-whisper is an optimized CTranslate2 port of OpenAI's Whisper model; transcribes voice notes and audio entirely on-device. MIT.

How QI uses it. • Local, free speech-to-text for PC voice, LINE/Telegram voice notes, and the meeting room — device=cpu, compute_type=int8 (CUDA DLLs absent in this Windows Python) • Transcribes the audio into lyrics when no embedded tags are found (models base/small/medium/large) • Local on-device transcription of Telegram voice notes (small.en default). Nothing leaves the machine. • Transcribe a spoken name (teach pronunciation) and transcribe source-song lyrics for low-lyrics-change remixes

Applications. Claude Voice, M2V, OpenClaw, PersonalSong Studio, TubeScout

edge-tts (Microsoft neural voices)

Licence: GPL-3.0, GPL-3.0 (client) / MS voices, Mixed (MIT / Apache-2.0) · Used by 4 applications

What it is. A Python client for Microsoft's free online neural text-to-speech voices (e.g. en-US-AndrewNeural). High-quality multilingual TTS at no cost. GPL-3.0 client.

How QI uses it. • Free neural voices for ja, fr, ru, de where Kokoro-82M lacks a usable voice • Claude's MALE voice — en-US-AndrewNeural (EN) / pt-BR-AntonioNeural (deferred); played via ffmpeg • edge-tts Microsoft neural voices (online, default); Piper local offline fallback; dots.tts GPU clone (parked). • Microsoft neural voices for native-language spoken dedications (8 languages, male/female, tone/style prosody)

Applications. AvatarStudio, Claude Voice, OpenClaw, PersonalSong Studio

Local open-weight LLMs (Qwen/Gemma/DeepSeek/etc.)

Licence: Derived from base model (e.g. Qwen2.5), Free tier / local, Mixed open weights / MIT runtime · Used by 3 applications

What it is. Open-weight language models (Qwen2.5-coder, sqlcoder, Gemma, DeepSeek-R1, GPT-OSS, granite, etc.) run locally through Ollama for reasoning, NL→SQL and chat at zero marginal cost. Mostly Apache-2.0 / MIT weights.

How QI uses it. • Specialized Ollama variant built from AutoPDFMapper.Modelfile; knows the 3 field kinds + library convention. • Local, free, private inference at 127.0.0.1:11434 — first in the failover chain (DeepSeek-R1 8B is primary) • Summary chain pinned to Groq→Cloudflare→Gemini; Discover reasoner defaults to Groq, switchable to fully-local gemma4

Applications. AutoPDF, Naya, TubeScout

PyTorch (CUDA)

Licence: BSD, BSD-3 · Used by 3 applications

What it is. The core deep-learning tensor framework with GPU (CUDA) acceleration that runs the local image/video/audio/speech models. BSD.

How QI uses it. • Backs Kokoro TTS in the Windows venv; render models run in their own WSL2 conda envs • GPU tensor backend for SDXL/SVD/Whisper; cu128 build required for the RTX 5080 (installed before other deps) • Tensor runtime for ACE-Step / Demucs / Seed-VC / Whisper / SDXL on the local GPU

Applications. AvatarStudio, M2V, PersonalSong Studio

Kokoro-82M (offline TTS)

Licence: Apache 2.0, Apache-2.0 / open · Used by 2 applications

What it is. A compact offline neural text-to-speech model — primary local voice engine for several languages, no network needed. Apache-2.0.

How QI uses it. • Offline neural TTS for en-US, en-GB, pt-BR, es, it, zh — primary voice engine • Planned spine for low-latency, interruptible voice and the Zoom-like screen-share UI (Phases 1-2)

Applications. AvatarStudio, Claude Voice

librosa / audio analysis

Licence: BSD / MIT / BSD / ISC, ISC · Used by 2 applications

What it is. A Python library for music/audio analysis — duration, MFCC features and structural segmentation to place scene boundaries. ISC.

How QI uses it. • Duration + MFCC/agglomerative structural segmentation to place scene boundaries • WAV write (bypasses TorchCodec), audio mixing/overlay/export, signal math, pitch (f0) for gender detection

Applications. M2V, PersonalSong Studio

Stable Diffusion XL / SVD (diffusers)

Licence: Apache 2.0, Apache 2.0 / model card, CreativeML OpenRAIL++-M, Stability AI Community · Used by 2 applications

What it is. Open image/video generative models (SDXL, SDXL-Turbo, Stable Video Diffusion) run via HuggingFace 'diffusers' to paint stills and animate them locally on GPU. Apache-2.0 code / model cards.

How QI uses it. • Generates one still per scene (1024x576, 30 steps, fp16) • Animates each still into 25-frame motion video (img2vid-xt, fp16, model CPU offload) • Loads and runs the SDXL/SVD pipelines and IP-Adapter weights • Optional AI-painted cover via StableDiffusionXLPipeline in a subprocess (own VRAM); PIL fallback on any error

Applications. M2V, PersonalSong Studio

ACE-Step (singing model)

Licence: Apache 2.0 (model/code) · Used by 1 application

What it is. A local diffusion model that generates full songs with sung vocals from lyrics plus style tags. Apache-2.0.

How QI uses it. • Local diffusion singing model — full songs with SUNG vocals from lyrics + style tags; bf16 on the local GPU; audio2audio for remix

Applications. PersonalSong Studio

ComfyUI

Licence: GPL-3.0 · Used by 1 application

What it is. A node-graph engine for Stable Diffusion pipelines, driven over its HTTP API to generate frames. GPL-3.0.

How QI uses it. • External SDXL frame generator the standalone track drives over urllib to produce scene_##.png (not yet run)

Applications. M2V

Demucs (stem separation)

Licence: MIT · Used by 1 application

What it is. Meta's open-source model that splits a track into stems (vocals / instrumental) for karaoke and voice-clone reference. MIT.

How QI uses it. • Two-stem (vocals / instrumental) split — backing tracks, karaoke, voice-clone reference, gender detection

Applications. PersonalSong Studio

IP-Adapter (character consistency)

Licence: Apache 2.0 · Used by 1 application

What it is. A conditioning adapter that locks SDXL to a reference image so the same face/style persists across generated scenes. Apache-2.0.

How QI uses it. • Conditions SDXL on a reference photo so the same faces/style persist across every scene (ip-adapter_sdxl.bin, scale configurable)

Applications. M2V

NotebookLM (vision-driven)

Licence: Commercial (free) · Used by 1 application

What it is. Google's NotebookLM driven through a browser session with vision models as a durable archive target. Commercial (Google).

How QI uses it. • Durable archive for Kaze/Yubin output; UI driven via Chrome session + qwen3-vl vision (no Google API).

Applications. OpenClaw

OpenCV (computer vision)

Licence: Apache 2.0 · Used by 1 application

What it is. The standard open-source computer-vision library, used here for face/eye detection and sharpness scoring to pick the best portrait frame. Apache-2.0.

How QI uses it. • Best-portrait-frame scoring: face/eye detection + Laplacian sharpness over sampled frames

Applications. AvatarStudio

Piper / dots.tts / Windows SAPI (TTS fallback)

Licence: Proprietary (OS) · Used by 1 application

What it is. Local/offline text-to-speech fallbacks: Piper (neural, MIT), Windows SAPI System.Speech (built-in 'David' voice) when network voices are unavailable.

How QI uses it. • Offline zero-dependency fallback voice (male 'David') when edge-tts/network is unavailable

Applications. Claude Voice

rembg (background removal)

Licence: MIT · Used by 1 application

What it is. An open-source model/tool that removes image backgrounds (cutout) for compositing. MIT.

How QI uses it. • Background removal (cutout) and solid-color / uploaded-image compositing on the portrait

Applications. AvatarStudio

Seed-VC (voice conversion)

Licence: GPL-3.0 (upstream) · Used by 1 application

What it is. A zero-shot voice-conversion model that re-sings or re-voices audio in a target timbre, run in an isolated environment. MIT.

How QI uses it. • Zero-shot voice conversion — re-sing/re-voice in a target timbre; runs in its OWN isolated venv as a subprocess

Applications. PersonalSong Studio

Talking-head render engines (Hallo2/LivePortrait/SadTalker/etc.)

Licence: Apache 2.0 (research), MIT (research), Mixed (research) · Used by 1 application

What it is. Audio- or video-driven avatar animation models (Hallo2, LivePortrait, MuseTalk, SadTalker, Wav2Lip, video-retalking) that make a still portrait speak/move; run in WSL2. Research/MIT licenses.

How QI uses it. • Driving-video motion transfer (~30s render); lips not audio-driven • Audio-driven talking-head diffusion render; lips match the waveform; slow • Optional lip-sync correction pass on top of any rendered video • Alternate render engines, wrappers present but disabled until their WSL2 envs are installed

Applications. AvatarStudio

VAD & wake-word (silero/web rtc/openWakeWord)

Licence: Apache-2.0, MIT / MIT · Used by 1 application

What it is. Voice-activity detection and wake-word models that decide when the user is speaking and when to start listening, for hands-free voice loops. MIT / Apache-2.0.

How QI uses it. • Hands-free turn endpointing in the realtime loop; energy VAD is the built-in default (no extra deps), silero/web rtc optional • Optional 'Hey Claude' gating as an alternative to always-on VAD (listening.mode=wake)

Applications. Claude Voice

AI — Cloud & Hosted Models

Anthropic Claude (API / SDK)

Licence: Anthropic (commercial), Commercial (Anthropic), Commercial API, MIT (client), Mixed (mostly paid; OpenRouter :free tier), Paid / Free tier, Paid API · Used by 9 applications

What it is. Anthropic's Claude family (Opus / Sonnet / Haiku) accessed via the Messages API and official SDK — the frontier-grade reasoning option, used where local models aren't enough. Commercial API.

How QI uses it. • Per-agent model tiers: Opus (architect, deep design), Sonnet (builder, inspector, librarian), Haiku (ops, scout, scribe, tester). • Optional brains: Anthropic messages API (claude-opus-4-8, disabled) and OpenRouter :free models (key from env or shared maia.db) • Default active chat vendor via the anthropic SDK; ClaudeVendor (default model claude-sonnet-4-6) • Alternative classifier/chat via /messages endpoint

Applications. Claude Manager, Claude Voice, CogniBase, EasyFlow, Gamez, LotteryWiz, MapSnap, Naya, QI Hive

OpenAI / ChatGPT (API)

Licence: Commercial, Paid API · Used by 4 applications

What it is. OpenAI's GPT models (e.g. gpt-4o-mini, gpt-5.x) via the OpenAI API or OAuth Codex profile — an alternative cloud reasoning/synthesis provider. Commercial API.

How QI uses it. • Concrete adapters present, active:false pending keys; OpenAI also supplies embeddings (text-embedding-3-*) • Alternative classifier/chat via /chat/completions; custom baseUrl supported • gpt-4o-mini via openai_codex adapter — dispatch + synthesis fallback • gpt-5.4 available through an OAuth profile (rennesan@gmail.com) for select agent work.

Applications. CogniBase, EasyFlow, NEXUS, OpenClaw

Cloudflare Workers AI

Licence: Commercial (free tier), Free tier · Used by 3 applications

What it is. Cloudflare's serverless AI inference (e.g. Qwen 7B) on a generous free tier — the default low-cost cloud LLM for several tools. Commercial (free tier).

How QI uses it. • Primary post-generation LLM — @cf/qwen/qwen2.5-7b-instruct-fast (free tier). Needs CF_ACCOUNT_ID + CF_AI_TOKEN; not yet configured • llama-3.3-70b-fp8-fast dispatch + scout synthesis (free tier) • Kaze digest primary provider — @cf/meta/llama-3.1-8b-instruct on the free tier, with automatic Ollama fallback.

Applications. MQ, NEXUS, OpenClaw

OpenRouter (model gateway)

Licence: Commercial API, Free tier, Mixed (mostly paid) · Used by 3 applications

What it is. An OpenAI-compatible gateway that proxies many model vendors behind one API (including free-tier models), used as a flexible cloud escape hatch. Commercial API.

How QI uses it. • Optional frontier-model access, gated by the per-profile data_policy guardrail. • Registered adapters, off by default — enable by adding a key (generic_openai covers OpenAI-compatible vendors) • Free-tier cloud models: Cloudflare Qwen1.5 7B, MiniMax M2.5, GPT-OSS 120B

Applications. MapSnap, NEXUS, Naya

D-ID / HeyGen (avatar SaaS)

Licence: Commercial SaaS · Used by 1 application

What it is. Commercial cloud APIs that render talking-photo / talking-head avatar videos (upload → generate → poll → download) for production-grade output. Commercial SaaS.

How QI uses it. • Cloud talking-photo rendering (upload -> generate -> poll -> download); for production polish

Applications. AvatarStudio

DeepEval (LLM evaluation)

Licence: Apache 2.0 · Used by 1 application

What it is. An open-source framework for scoring LLM output quality (G-Eval clarity/accuracy/depth). Apache-2.0.

How QI uses it. • Structured Clarity/Accuracy/Depth scoring via the NexusJudge bridge; default scorer mode

Applications. NEXUS

DeepL (translation)

Licence: Paid / free-tier API · Used by 1 application

What it is. A high-quality machine-translation API (preferred JA→EN engine) with free and pro tiers. Commercial API.

How QI uses it. • Preferred JA->EN engine when DEEPL_API_KEY is set (auto-detects free vs pro host) — currently unconfigured

Applications. AkiyaScout

Google Gemini

Licence: Free tier / Paid · Used by 1 application

What it is. Google's Gemini models (gemini-2.0-flash) used as a default classifier/chat assistant; free tier plus paid. Commercial API.

How QI uses it. • Default classifier + chat assistant (gemini-2.0-flash generateContent). User supplies key

Applications. EasyFlow

Groq

Licence: Free tier · Used by 1 application

What it is. A cloud inference provider serving open models (Llama-3.3-70B) at very high speed on a free tier — used as a fast judge/summary provider. Commercial (free tier).

How QI uses it. • llama-3.3-70b-versatile — fast inference + default judge provider

Applications. NEXUS

Mistral

Licence: Free tier · Used by 1 application

What it is. Mistral AI's hosted models (mistral-small) used as a dispatch provider. Commercial API.

How QI uses it. • mistral-small-latest dispatch provider

Applications. NEXUS

tiktoken

Licence: MIT · Used by 1 application

What it is. OpenAI's fast tokenizer library, used to count tokens and size RAG chunks. MIT.

How QI uses it. • Token counting / chunk sizing for the RAG pipeline

Applications. CogniBase

Document, PDF, Office & Media Processing

FFmpeg

Licence: HPND / BSD, LGPL/GPL · Used by 4 applications

What it is. The universal open-source audio/video toolkit — transcode, concat, mux, scale, subtitle-burn, and probe (ffprobe). Used everywhere media is produced. LGPL/GPL.

How QI uses it. • Audio mux into renders, frame extraction, duration probe, scene concatenation • mp3->m4a (LINE), mp3->OGG/Opus (Telegram), 16k mono WAV for STT, headless playback; resolved via util.py (PATH/WinGet glob) • Image handling for SDXL output and frame export; imageio-ffmpeg bundles an FFmpeg binary for diffusers export_to_video • Concatenates clips, muxes the original audio, scales/loops to duration, and burns SRT subtitles via the subtitles filter

Applications. AvatarStudio, Claude Voice, M2V, PersonalSong Studio

lxml / selectolax / BeautifulSoup (HTML/XML parsing)

Licence: BSD, MIT · Used by 3 applications

What it is. HTML/XML parsers for scraping and for reading export packages: lxml (BSD), selectolax (MIT), BeautifulSoup (MIT).

How QI uses it. • Fast HTML parsing for the live portal adapters (section.property on at-home, div.property_unit on SUUMO) • XML parsing of OnBase Studio .expk export packages in the discovery pipeline • Parses lottonumbers.com year archives to backfill Fantasy 5 history (official API gives only the latest)

Applications. AkiyaScout, CogniBase, LotteryWiz

openpyxl

Licence: MIT · Used by 3 applications

What it is. A Python library to read and write Excel .xlsx workbooks (schema exports, report output). MIT.

How QI uses it. • Spreadsheet (.xlsx) export for report outputs • Styled .xlsx play-sheet generation (gold header, borders, metadata); CSV via stdlib csv • Reads .xlsx schema/relation exports (e.g. OnBase IST schema workbooks).

Applications. CogniBase, LotteryWiz, MapSnap

PDF text extraction (PyMuPDF/Poppler/PyPDF2/pdfminer)

Licence: AGPL / commercial, BSD / MIT, GPL-2.0-or-later · Used by 3 applications

What it is. Libraries that pull text and word positions out of PDFs for search, zones and RAG: PyMuPDF/fitz (AGPL/commercial), Poppler pdftotext (GPL), PyPDF2 / pdfminer.six (BSD/MIT).

How QI uses it. • Text + word-bbox extraction for zones, regex, anchors and tagged mapping. • PDF text/content extraction from OnBase document binaries • Optional PDF text extraction for the document library (graceful fallback if absent).

Applications. AutoPDF, CogniBase, MapSnap

Microsoft Office COM / pywin32

Licence: Commercial (Microsoft), PSF-style · Used by 2 applications

What it is. Automation of installed Word/Excel/PowerPoint via COM (pywin32) for high-fidelity document conversion/export. Commercial Office; pywin32 PSF.

How QI uses it. • Optional: Word/Excel/PowerPoint and styled Markdown conversion via COM automation. • Optional PDF export via Word COM automation

Applications. AutoPDF, Naya

mutagen (audio tags)

Licence: BSD / GPL-2.0, GPL-2.0 · Used by 2 applications

What it is. A Python library that reads/writes audio metadata (ID3, lyric tags) for finished MP3s. GPL.

How QI uses it. • Audio I/O; mutagen also reads ID3/USLT/SYLT lyric tags and song metadata • ID3 tagging of finished MP3s for Plex playback

Applications. M2V, PersonalSong Studio

soundfile / pydub / scipy / numpy (audio I/O)

Licence: BSD, BSD / BSD / BSD · Used by 2 applications

What it is. Audio read/write, mixing, resampling and signal math for TTS chunk assembly and song pipelines. BSD/MIT.

How QI uses it. • Read/write WAV (24 kHz), concatenate Kokoro chunks, pitch-shift via resampling • Microphone capture, WAV write, and RMS energy computation for the VAD / barge-in loop

Applications. AvatarStudio, Claude Voice

Ghostscript

Licence: AGPL-3.0-or-later · Used by 1 application

What it is. The standard open-source PostScript/PDF interpreter, called as a separate process to render and convert PDFs. AGPL-3.0.

How QI uses it. • Core PDF rendering / conversion; bundled and called as a separate process (gswin64c.exe).

Applications. AutoPDF

matplotlib + pandas

Licence: PSF / BSD · Used by 1 application

What it is. matplotlib renders charts; pandas shapes tabular data — used for reports and analytics. BSD.

How QI uses it. • Chart rendering and tabular shaping for reports and the future Canvas

Applications. CogniBase

MoviePy

Licence: MIT · Used by 1 application

What it is. A Python video-editing library (Ken Burns, crossfades, captions) used for a proof-of-concept renderer. MIT.

How QI uses it. • Standalone proof-of-concept renderer: Ken Burns zoom, crossfades, lower-third captions, placeholder fallback (make_forro_video.py)

Applications. M2V

NAPS2 (scanner bridge)

Licence: GPL-2.0-or-later · Used by 1 application

What it is. An open-source document-scanning tool (TWAIN/ESCL) invoked via its console to capture from scanners. GPL.

How QI uses it. • TWAIN / ESCL scanner bridge behind Invoke-ScanCapture; native WIA used otherwise.

Applications. AutoPDF

NumPy / SciPy

Licence: BSD · Used by 1 application

What it is. Foundational numerical-computing libraries for arrays, sampling and signal math. BSD.

How QI uses it. • Exact multinomial sampling for the Brutal Ledger random control (microseconds); pure-Python fallback if absent

Applications. LotteryWiz

PDFtk (pdftk-java)

Licence: GPL-2.0-only · Used by 1 application

What it is. A Java tool for PDF page operations — split, merge, rotate, stamp. GPL.

How QI uses it. • PDF page operations / manipulation used by the splitter.

Applications. AutoPDF

Pillow (PIL)

Licence: HPND/MIT-CMU · Used by 1 application

What it is. The standard Python imaging library — generate, composite and export images (cover art, frames). HPND/MIT-style.

How QI uses it. • Procedural gradient cover art — genre palette, glow, music note, name/occasion. Always available, instant

Applications. PersonalSong Studio

qrcode

Licence: BSD · Used by 1 application

What it is. A Python library that generates scannable QR codes (e.g. printable cards linking to a play page). BSD.

How QI uses it. • Printable greeting card with a scannable QR linking to the song's play page (tunnel/LAN-aware)

Applications. PersonalSong Studio

RapidOCR (ONNX)

Licence: Apache-2.0 (planned) · Used by 1 application

What it is. A planned second OCR engine (PP-OCR models) running via ONNX Runtime in .NET — no Python dependency. Apache-2.0.

How QI uses it. • Planned second OCR engine (PP-OCRv5/v6) via .NET ONNX binding; no Python; CPU/CUDA selectable.

Applications. AutoPDF

Tabula (tabula-java)

Licence: MIT · Used by 1 application

What it is. A Java library that extracts tables from PDFs into structured rows (PDF → Excel). MIT.

How QI uses it. • True PDF -> Excel table extraction.

Applications. AutoPDF

Tesseract OCR

Licence: Apache-2.0 · Used by 1 application

What it is. The leading open-source OCR engine; adds a searchable text layer to scanned/image-only PDFs in many languages. Apache-2.0.

How QI uses it. • Default OCR engine; produces a searchable text layer for scanned / image-only PDFs; multi-language.

Applications. AutoPDF

ytmusicapi

Licence: MIT · Used by 1 application

What it is. An unofficial client for YouTube Music (upload/playlist management). MIT.

How QI uses it. • YouTube Music browser-header auth + upload/playlist management

Applications. PersonalSong Studio

zipfile + xml (stdlib)

Licence: PSF · Used by 1 application

What it is. Python's built-in ZIP and XML handling, used to unpack OnBase Studio .spk/.expk packages (ZIP+XML). PSF.

How QI uses it. • Unpacks OnBase .spk/.expk Studio packages (ZIP+XML) to extract DocType / keyword definitions.

Applications. MapSnap

Service Management, Tunnels & Scheduling

NSSM (Windows service manager)

Licence: Cloudflare / open source, Cloudflare ToS / NSSM public domain, Mixed (GPL / public domain / MS), Open source / Cloudflare, Public domain, Public domain (nssm) / Windows, Public domain / Cloudflare · Used by 18 applications

What it is. The Non-Sucking Service Manager runs any executable as an auto-start Windows service with log rotation — the QI standard for keeping apps and tunnels alive 24/7 (services prefixed QI_). Public domain.

How QI uses it. • QI_AutoPDFTunnel (auto-start) exposes PIN-gated public access; QI_AutoPDF manual placeholder; tunnel_manager.py + nssm.exe. • QI_AvatarStudio demand-start service (AppDirectory C:\1-AI\APPS\AvatarStudio, port 7862) • QI_ NSSM services are supervised (not owned) here; QI_ClaudeSelfAudit and QI_ClaudeVoiceMeeting_8AM scheduled tasks drive monthly + daily automation. • QI_ClaudeVoice* services (Line/Telegram/Tunnel/Control) + tunnel claudevoice.quiddityinnovations.com + daily meeting task + hourly bridge check

Applications. AutoPDF, AvatarStudio, Claude Manager, Claude Voice, CogniBase, CypherMiner, Gamez, LotteryWiz, M2V, MQ, MapSnap, NEXUS, Naya, OpenClaw, PersonalSong Studio, QI Brain, QI Hive, TubeScout

APScheduler

Licence: MIT · Used by 5 applications

What it is. An in-process Python job scheduler (intervals, cron, daily digests) with timezone support. MIT.

How QI uses it. • BackgroundScheduler (Asia/Tokyo): interval ingest, 08:00 daily digest, per-profile cron digests — POC-grade, no Redis/Celery • BackgroundScheduler + CronTrigger for recurring scan / dedup tasks (wired; no jobs registered yet) • BackgroundScheduler re-fetching + re-scoring every refresh_minutes (15) • BackgroundScheduler running per-topic Scout cycles, scheduled prompt jobs, and synthesis keep-warm

Applications. AkiyaScout, FileHQ, Gamez, NEXUS, Naya

Cloudflare Tunnel (cloudflared)

Licence: Apache-2.0 (client), Commercial (free), Proprietary (free tier) · Used by 3 applications

What it is. Cloudflare's tunnel client exposes a localhost port at a stable public HTTPS URL without opening firewall ports — how QI apps reach the public internet. Apache-2.0 client.

How QI uses it. • Named tunnel exposing localhost:9876 as https://mapsnap.quiddityinnovations.com; quick tunnel fallback on customer boxes. • Permanent named tunnel publishing the Kaze news page at kaze.quiddityinnovations.com. • Static named tunnels — one permanent subdomain per app on quiddityinnovations.com

Applications. MapSnap, OpenClaw, QI Hive

Windows Task Scheduler

Licence: OS, OS built-in, OS component · Used by 3 applications

What it is. The built-in Windows scheduler (schtasks) that runs pipelines and jobs on a clock when an in-process scheduler isn't wanted. OS component.

How QI uses it. • Runs the Fumi pipelines on cadence (FumiMaiaWeekly Sat, FumiQIDaily Mon-Fri). Setup script ready; jobs not registered • Registers deferred / recurring scan jobs under the Naya\ task prefix • QI_TubeScout_AM (07:00) + QI_TubeScout_PM (19:00) run shared.cycle daily; no in-process scheduler

Applications. MQ, Naya, TubeScout

WSL / WSL2 (Ubuntu)

Licence: Open source, Ubuntu (Canonical) / WSL (MS) · Used by 2 applications

What it is. Windows Subsystem for Linux — a real Ubuntu environment inside Windows that hosts Linux-only render engines, gateways and Ollama. Microsoft, free.

How QI uses it. • Runs all render engines + ffmpeg; driven from Windows via 'wsl -d Ubuntu-24.04 bash -c'; Windows paths translated to /mnt/c • Linux environment hosting the gateway, agents, schedulers, and Ollama under Windows.

Applications. AvatarStudio, OpenClaw

GitHub Actions

Licence: Proprietary (CI) · Used by 1 application

What it is. GitHub's CI/CD service that builds and packages apps on Windows/macOS/Linux runners. Proprietary (free tier).

How QI uses it. • Builds the windowed app on Windows + macOS + Linux runners and uploads all three artifacts

Applications. Gamez

PyInstaller

Licence: GPL w/ exception · Used by 1 application

What it is. Bundles a Python app plus its dependencies into a single self-contained executable for distribution. GPL-with-exception.

How QI uses it. • Bundles the proxy + HTML + config into a single self-contained app (windowed + onedir builds)

Applications. Gamez

setuptools + Docker

Licence: MIT / Apache 2.0 · Used by 1 application

What it is. setuptools packages Python projects (pip installable); Docker containerizes them for reproducible deployment. MIT / Apache-2.0.

How QI uses it. • pip editable install (cognibase-server / cognibase-rag scripts) and Docker/compose with bind-mounted state for full portability

Applications. CogniBase

Win32 process control (CIM/Stop-Process)

Licence: Windows · Used by 1 application

What it is. WMI/CIM and Stop-Process used by the restart guard to map and kill orphaned process trees. OS component.

How QI uses it. • The restart guard maps the full process tree, walks ancestors/descendants, and kills orphaned MCP trees while protecting QI_ services and its own chain.

Applications. Claude Manager

External Integrations & Data Sources

feedparser (RSS/Atom)

Licence: BSD · Used by 3 applications

What it is. A Python library that parses RSS and Atom feeds for news/source ingestion. BSD.

How QI uses it. • RSS/Atom parsing for Kaze's travel / photo / AI source feeds • RSS/Atom parsing for Scout RSS sources (8,500+ items collected) • RSS/Atom parsing for Kaze; combined with ThreadPoolExecutor fetch and SequenceMatcher de-dup.

Applications. MQ, NEXUS, OpenClaw

LINE Messaging API

Licence: Apache-2.0, Commercial (LINE), Commercial (free) · Used by 3 applications

What it is. LINE's official bot platform (webhooks + push) for chat delivery and alerting; the line-bot-sdk handles signed webhooks and rich replies. Commercial (free tier).

How QI uses it. • Outbound alerting to Renne via broadcast — no stored userId; token from C:\OC\secrets\tasuke-line.env. • Signed webhook parsing + messaging/blob APIs for text, audio, and video replies on :8721 • Inbound/outbound messaging. Telegram polled by the gateway; LINE accounts for tasuke/naya/kaze.

Applications. Claude Manager, Claude Voice, OpenClaw

Gmail API / IMAP-SMTP

Licence: Commercial (free), Free (Google quotas apply) · Used by 2 applications

What it is. Programmatic Gmail access — the Gmail REST API (labels, filters, modify) and classic IMAP/SMTP for mailbox read/send. Free within Google quotas.

How QI uses it. • Labels CRUD, messages.modify + batchModify, settings.filters CRUD, label visibility. Used by both the extension (REST via gmailFetch) and the Python tools (google-api-python-client) • Yubin mailbox access (imap.gmail.com:993 / smtp.gmail.com:587) for maia.quiddam@gmail.com via app password.

Applications. EasyFlow, OpenClaw

Google Workspace APIs (Apps Script / People / Drive)

Licence: Apache 2.0, Free (Google quotas apply) · Used by 2 applications

What it is. Google's REST APIs for Apps Script (push generated code), People (contact lookup) and related services. Free within quotas.

How QI uses it. • Contact-aware routing — sender contact lookup (contacts OAuth scope, people.googleapis.com) • projects.updateContent — push the generated classifier into the user's Apps Script project (script.projects scope, token_apps_script.json) • OAuth 2.0 InstalledAppFlow (youtube.readonly), token refresh, persisted to secrets/token.json

Applications. EasyFlow, TubeScout

Chrome / Edge Extension platform (Manifest V3)

Licence: Chrome extension APIs, Open standard, Open standard (Google), Store programs · Used by 1 application

What it is. The browser-extension runtime (side panel, content scripts, service worker) and store distribution (Chrome Web Store, Edge Add-ons) for in-browser tools. Open standard (Google).

How QI uses it. • Primary in-browser runtime — side panel, content script, background service worker; ships as Chrome and Edge builds • identity (OAuth), storage.local (state + keys + logs), alarms (scheduled classification), sidePanel, tabs, downloads, notifications, activeTab • 10-locales UI (en, zh_CN, fr, de, ja, ko, pt_BR, es, hi, ru); _locales/ message catalogs + shared/i18n + command-i18n + guide-i18n • Public listings (in review). CASA verification is the gating step for restricted Gmail scopes

Applications. EasyFlow

Facebook / Instagram / WhatsApp (Meta APIs)

Licence: Meta Platform Terms · Used by 1 application

What it is. Meta's Graph/Cloud APIs for publishing to Facebook Pages and (planned) Instagram and WhatsApp. Meta Platform Terms.

How QI uses it. • Page feed publishing at graph.facebook.com/v19.0/{page_id}/feed (Fumi). Blocked on Meta credentials • Planned — IG Content Publishing + WhatsApp Cloud API under the Maia Quiddam identity. No code yet

Applications. MQ

Lottery open-data feeds

Licence: Open data / official · Used by 1 application

What it is. Official and community lottery result feeds (data.ny.gov Socrata, Brazil Caixa, France FDJ, EuroMillions) for draw history. Open data.

How QI uses it. • Official US (Socrata), Brazil (Caixa), France (FDJ open data) feeds + EuroMillions community API for live results + jackpots

Applications. LotteryWiz

Map tiles & geocoding (OSM/Esri/GSI)

Licence: ODbL (free, usage policy), Tile-provider terms (attribution required) · Used by 1 application

What it is. Base-map tile providers (OpenStreetMap, Esri, CartoDB, Japan GSI/MLIT) and geocoders (Nominatim) for the property maps. Provider terms / attribution.

How QI uses it. • English-label, Japanese-label, satellite, topographic base layers + flood/landslide/tsunami/storm-surge hazard overlays • Municipality -> lat/long lookup (max ~1 req/sec, cached in geocache)

Applications. AkiyaScout

Plex (media server)

Licence: in-repo module · Used by 1 application

What it is. A personal media server; finished MP3s are tagged and copied into a Plex-watched folder and a section scan triggered. Commercial (free tier).

How QI uses it. • ID3 tag + copy to a Plex-watched folder + section scan; connection/section probes

Applications. PersonalSong Studio

praw (Reddit)

Licence: BSD · Used by 1 application

What it is. The Python Reddit API Wrapper, used to collect items from Reddit sources. BSD.

How QI uses it. • Reddit API client for Scout's Reddit source (4,700+ items collected)

Applications. NEXUS

Sports & odds data APIs (ESPN/API-Football/Kalshi/EA FC)

Licence: Community dataset, Free tier (paid plans), Public API (no key), Public endpoint (unofficial) · Used by 1 application

What it is. Public and free-tier sports and prediction-market data feeds: ESPN JSON, API-Football, Kalshi markets, EA FC ratings CSV. Mixed public/free-tier.

How QI uses it. • Fixtures, scores, 12-group standings, bracket flags — no API key • Default squad-strength source: real player 'overall' mapped to national teams (free, no key, no quota) • Deep per-player WC season stats for richer PPS — optional key, free tier ~100 req/day • Live World Cup market prices (implied probability + payout), fetched server-side to avoid CORS

Applications. Gamez

Telegram Bot API

Licence: Free API · Used by 1 application

What it is. Telegram's bot API used in outbound long-poll mode (getUpdates) — no inbound port or tunnel required. Free.

How QI uses it. • Outbound long-poll getUpdates loop (@Naya_qi_bot); owner-restricted; no inbound port

Applications. Naya

YouTube Data API & transcripts

Licence: MIT · Used by 1 application

What it is. YouTube Data API v3 (subscriptions, channels, uploads) plus caption/transcript fetching (youtube-transcript-api) for video intelligence. Apache-2.0 clients.

How QI uses it. • Caption fetch (instance .fetch()), languages en/pt/es/ja/fr); skip reasons vs. IP-block markers handled distinctly

Applications. TubeScout

yt-dlp

Licence: Unlicense · Used by 1 application

What it is. A powerful media downloader that fetches best-quality audio/video from YouTube and many sites. Unlicense (public domain).

How QI uses it. • Download bestaudio from a YouTube/other URL as WAV for voice-clone sources

Applications. PersonalSong Studio

Security, Auth & Cryptography

stdlib crypto (hmac / hashlib / secrets)

Licence: PSF, PSF (stdlib) · Used by 2 applications

What it is. Python's built-in cryptographic primitives for password hashing, session-token generation and file hashing (SHA-256). PSF.

How QI uses it. • Password hashing and session-token generation for login mode. • Streaming chunked file hashing for exact-duplicate detection

Applications. MapSnap, Naya

Authlib

Licence: BSD · Used by 1 application

What it is. A Python OAuth/OIDC library implementing the PKCE flow to an institution's identity provider. BSD.

How QI uses it. • OAuth2 PKCE flow to the institution's IdP for live OnBase access (M2 — config shape present, wiring pending)

Applications. CogniBase

Google OAuth2 (+ PKCE)

Licence: Open standard · Used by 1 application

What it is. The OAuth 2.0 / PKCE authorization flow for secure delegated access to Google APIs (Gmail, YouTube) without storing passwords. Open standard.

How QI uses it. • Extension: chrome.identity.getToken (Chrome) + launchWebAuthFlow (Edge). Tools: google-auth-oauthlib Flow with PKCE over localhost

Applications. EasyFlow

Hashing for dedup (xxhash / hashlib)

Licence: BSD-2 · Used by 1 application

What it is. Fast file fingerprinting for exact-duplicate detection — xxhash (non-cryptographic, BSD) and streaming SHA-256.

How QI uses it. • Fast non-cryptographic xxh64 fingerprinting — partial (64KB) and full-file hashes for exact-duplicate detection

Applications. FileHQ

sqlglot (SQL safety)

Licence: MIT · Used by 1 application

What it is. A SQL parser that builds an AST to enforce a read-only safety boundary (single SELECT, no writes). MIT.

How QI uses it. • Parses capability SQL into an AST to enforce the safety boundary: single read-only SELECT, anchor-scoped tables only, no DDL/PRAGMA/system tables

Applications. CogniBase

Web Crypto API

Licence: Browser built-in · Used by 1 application

What it is. The browser's native cryptography API (crypto.subtle) for in-page SHA digests. Web standard.

How QI uses it. • Native browser SHA-1/256/384/512 digests in the SHA tool

Applications. CypherMiner

Developer Tooling, Agents & Protocols

Git / GitHub

Licence: GPL, GPL / proprietary, GPLv2 · Used by 4 applications

What it is. The distributed version-control system and GitHub hosting — source of truth, nightly auto-sync, and drift detection across projects. GPLv2 (Git).

How QI uses it. • The supervisor reads git status/log per project for drift; the self-audit prunes ephemeral .claude/worktrees across 8 repos. • repo/ holds the curated, non-secret agent scripts + config templates (secrets excluded by design). • Poller shells out to git log --since to detect recent commits per project without an API push • Source at Quiddity-Innovations/QI-Hive; nightly auto-sync

Applications. Claude Manager, OpenClaw, QI Brain, QI Hive

Claude Code (Anthropic CLI / Agent SDK)

Licence: Commercial (Anthropic), Internal (QI), Subscription (Anthropic) · Used by 3 applications

What it is. Anthropic's agentic coding CLI and Agent SDK — the session host and orchestrator that runs the QI Hive sub-agents. Commercial (Anthropic).

How QI uses it. • The session host. Main-thread Claude is the orchestrator; sub-agents run via the Agent tool. This workspace is its management cockpit. • permissions (bypassPermissions) + SessionEnd/SubagentStop hook wiring; SessionStart/UserPromptSubmit hooks wired in the global

~/claude.config. • The genuine Claude via Renne's subscription token (CLAUDE_CODE_OAUTH_TOKEN); claude_cli backend, model 'sonnet', ~20s/turn, no metered API • usage_stats.py reads ~/claude/**/.jsonl for cost/token tracking — no API key

Applications. Claude Manager, Claude Voice, QI Hive

Model Context Protocol (MCP / FastMCP)

Licence: MIT, Open spec, n/a (QI internal) · Used by 3 applications

What it is. An open protocol (JSON-RPC over stdio) that exposes tools to AI clients like Claude; FastMCP is a Python server framework for it. MIT.

How QI uses it. • Live snapshot at session start (HTTP) and qi.* MCP tools (get_context, log_session, log_decision, search_memory). • mcp_server.py exposes 11 tools to Claude Desktop over stdio - drive Lottery Wiz on the subscription, no paid API • qi-brain stdio server exposes 12 Brain tools to Claude Code

Applications. Claude Manager, LotteryWiz, QI Hive

rapidfuzz / difflib (fuzzy matching)

Licence: MIT, PSF (stdlib) · Used by 2 applications

What it is. Fast string-similarity matching for de-duplication and near-duplicate detection — rapidfuzz (MIT) and stdlib difflib (PSF).

How QI uses it. • Fast fuzzy matching for Scout deduplication (85% similarity threshold) • Near-duplicate filename similarity for the fuzzy scanner

Applications. NEXUS, Naya

filesystem stdlib (os/shutil/pathlib)

Licence: PSF (stdlib) · Used by 1 application

What it is. Python's built-in filesystem modules for traversal, stat-based indexing and safe move/rename/delete. PSF.

How QI uses it. • Directory traversal, stat-based indexing, and move/rename/delete execution for the organizer

Applications. FileHQ

langdetect

Licence: Apache-2.0 · Used by 1 application

What it is. A language-detection library used to pick the right voice/locale per message. Apache-2.0.

How QI uses it. • Per-message language detection for voice selection (currently English-forced)

Applications. Claude Voice

Markdown front-matter agents

Licence: n/a (project files) · Used by 1 application

What it is. Each QI Hive sub-agent is defined as a Markdown file with YAML front-matter (name/model/tools) plus a role prompt. Internal (QI).

How QI uses it. • Each hive-* agent is a .md file with name/description/model/tools front-matter + a role prompt; loaded from .claude/agents/.

Applications. Claude Manager

Playwright / Pester (UI & API tests)

Licence: Apache-2.0 (Playwright) / MIT (pytest) · Used by 1 application

What it is. Playwright drives real-browser UI regression tests; Pester is the PowerShell test framework for API/AI tiers. MIT / Apache-2.0.

How QI uses it. • Tier 1/2 PowerShell API/AI tests (~27) and Tier 3 pytest + Playwright UI regression.

Applications. AutoPDF

send2trash

Licence: BSD · Used by 1 application

What it is. A cross-platform library that sends files to the Recycle Bin instead of hard-deleting. BSD.

How QI uses it. • Recycle-bin-first deletion (graceful fallback if missing)

Applications. Naya

QI Internal Platform Services

NEXUS (multi-AI synthesis)

Licence: QI internal, Quiddity Innovations (internal) · Used by 3 applications

What it is. QI's multi-provider AI synthesis service (port 8010) that fans a prompt across many models and merges the answers; consumed by sibling projects for summarization and scoring. Internal (QI).

How QI uses it. • Declared consumes_from_ecosystem dependency: LLM recommendations for file analysis (not yet wired in code) • Optional multi-AI synthesis via POST :8010 /synthesize; off by default • All summarization + implement-worthiness scoring + recommender reasoning via POST /synthesize; OPML ingest via /scout/import/opml. No LLM runs inside TubeScout

Applications. FileHQ, Naya, TubeScout

Maia / Naya shared modules

Licence: QI internal, Quiddity Innovations (internal) · Used by 2 applications

What it is. Reused database/context/language modules from the Maia bot, repointed so sibling bots (Naya) run on their own DB. Internal (QI).

How QI uses it. • FileHQ reads personal.default_scan_drive from naya.db and is launched + consumed by the Naya bot; FileHQ has no separate identity layer • Reused from Maia; naya_db.py repoints maia_db.DB_FILE to naya.db so they run against Naya's own data

Applications. FileHQ, Naya

QI Brain (memory + knowledge graph)

Licence: Internal (QI), QI internal · Used by 2 applications

What it is. QI's shared 'brain' — a SQLite + ChromaDB service (port 9011) that stores decisions, features, sessions, project state and a documentation knowledge graph, exposed over HTTP and via the qi-brain MCP server. Internal (QI).

How QI uses it. • Scene AI split (NEXUS :8010), session/feature logging (Brain :9011), output surfacing (dashboard :8600) • Brain receives implement-worthy features; Kaze digest consumes the curated OPML; Hive supplies the live NEXUS tunnel URL

Applications. AvatarStudio, TubeScout

QI_Elevate broker

Licence: Internal (QI), QI internal · Used by 2 applications

What it is. A LocalSystem daemon that executes a whitelist of admin commands so agents can control services without interactive UAC. Internal (QI).

How QI uses it. • LocalSystem daemon executing whitelisted admin commands for autonomous service control • install_service.py registers the NSSM service through qi_elevate_client.run_elevated() (the QI standard for elevated ops)

Applications. QI Hive, TubeScout

OpenClaw (agent gateway)

Licence: Open source (npm package) · Used by 1 application

What it is. A self-hosted agent runtime (npm gateway) that polls Telegram, runs the active LLM agent and executes shell tools under WSL. Internal (QI).

How QI uses it. • Self-hosted agent runtime: polls Telegram, runs the active LLM agent, executes shell via TOOLS.md flows + exec approvals.

Applications. OpenClaw

qi_registry.json (ecosystem registry)

Licence: Internal (QI) · Used by 1 application

What it is. The single source of truth for every QI project's ports, paths, status and integration contracts — read at bootstrap for CORS and port allocation. Internal (QI).

How QI uses it. • Compliance checker + compliant-project scaffolder enforcing the Six Laws

Applications. QI Hive

Other / Project-Specific Components

AutoPDF-Server.ps1

Licence: Proprietary (Quiddity Innovations) · Used by 1 application

How QI uses it. • Single-file engine: convert, split, extract, index, OnBase XML, AI proxy, config import/export, scheduler.

Applications. AutoPDF

Browser print-to-PDF

Licence: Browser-native · Used by 1 application

How QI uses it. • The signed report is produced via `window.print()` and the browser's 'Save as PDF'; on-screen it renders as stacked 8.5x11 pages. No PDF library bundled.

Applications. Digitization Cost Tool

Built-in glossary stub

Licence: In-house · Used by 1 application

How QI uses it. • Zero-dependency real-estate term translator — deterministic, demo-safe last resort

Applications. AkiyaScout

file:// · web server · SharePoint embed

Licence: — · Used by 1 application

How QI uses it. • Runs from a local file, a USB/shared drive, an HTTPS web server, or an embedded SharePoint page (which requires the web-hosted URL, since SharePoint blocks page JS).

Applications. Digitization Cost Tool

OnBase XML Index DIP

Licence: n/a (output format) · Used by 1 application

How QI uses it. • Hyland-schema XML index output (`Write-IndexXml`); `ConvertTo-XmlName` for safe element names.

Applications. AutoPDF

smtplib (stdlib)

Licence: PSF (stdlib) · Used by 1 application

How QI uses it. • Optional local SMTP digest delivery (off by default; needs `SMTP_PASSWORD` env)

Applications. AkiyaScout

Appendix — Application coverage by category

A quick cross-reference: which categories each application draws on. A dot means the application uses at least one technology from that category.

Application	Categories used
AkiyaScout	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C14
AutoPDF	C1, C3, C5, C6, C8, C9, C12, C14
AvatarStudio	C1, C2, C4, C6, C7, C8, C9, C13
Claude Manager	C1, C5, C7, C9, C10, C12
Claude Voice	C1, C2, C4, C5, C6, C7, C8, C9, C10, C12
CogniBase	C1, C2, C4, C5, C6, C7, C8, C9, C11
CypherMiner	C1, C2, C3, C4, C9, C11
Digitization Cost Tool	C1, C3, C14
EasyFlow	C1, C2, C6, C7, C10, C11
FileHQ	C1, C2, C4, C5, C9, C11, C12, C13
Gamez	C1, C2, C3, C4, C5, C6, C7, C9, C10
LotteryWiz	C1, C2, C4, C5, C6, C7, C8, C9, C10, C12
M2V	C1, C2, C4, C5, C6, C8, C9
MQ	C1, C2, C4, C5, C6, C7, C9, C10
MapSnap	C1, C5, C6, C7, C8, C9, C11
NEXUS	C1, C2, C4, C5, C6, C7, C9, C10, C12
Naya	C1, C2, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13
OpenClaw	C1, C5, C6, C7, C9, C10, C12, C13
PersonalSong Studio	C1, C2, C4, C6, C8, C9, C10
QI Brain	C1, C2, C4, C5, C6, C9, C12
QI Hive	C1, C2, C3, C4, C5, C6, C7, C9, C12, C13
TubeScout	C1, C2, C4, C5, C6, C9, C10, C13

C1 = Languages & Runtimes

C2 = Web & API Frameworks / Servers

C3 = Frontend, UI & Styling

C4 = Data Validation, HTTP Clients & Concurrency

C5 = Databases, Vector Stores & On-Disk State

C6 = AI — Local Inference (LLMs, Embeddings, Speech, Vision, Media)

C7 = AI — Cloud & Hosted Models

C8 = Document, PDF, Office & Media Processing

C9 = Service Management, Tunnels & Scheduling

C10 = External Integrations & Data Sources

C11 = Security, Auth & Cryptography

C12 = Developer Tooling, Agents & Protocols

C13 = QI Internal Platform Services

C14 = Other / Project-Specific Components